

Setup (2 minutes): one laptop. From the bCourses Class 3 page download “CPS.xlsx”: 139,931 civilian, non-institutionalized adults from the March 2019 Current Population Survey — the monthly survey of about 60,000 households behind the official U.S. unemployment rate. Use any AI that can run code. Today the AI computes outputs; your team audits the definitions, the interpretation, and the decision.

**1. What is the unemployment rate?**

Give your AI the dataset and this exact request: “Compute the unemployment rate and a 95% confidence interval for it. Show exactly which Employment Status categories you counted as unemployed, which as employed, and what denominator you used.”

- a. Getting this right requires definitional choices, not just arithmetic. Check the AI’s category mapping against the seven Employment Status categories: who did it count as unemployed? As employed? Who is excluded from the denominator entirely? Did it get it right?

*Solution: Unemployed = “Unemployed, experienced worker” (3,241) + “Unemployed, new worker” (338) = 3,579. Employed = “At work” (80,730) + “Has job, not at work last week” (2,829) = 83,559 — the vacationing/sick-leave category is the common miss. Everyone “NILF” (retired, unable to work, other) is excluded from the denominator. Rate =  $3,579/87,138 = 4.1\%$ . AIs often get this right — the point is you cannot know without demanding the mapping.*

- b. A colleague computes  $3,579 \div 139,931 = 2.6\%$  and says “unemployment is even lower than reported!” In one or two sentences: why is that not the unemployment rate?

*Solution: The denominator includes 52,793 people not in the labor force — retirees, students, people unable to work — who are neither employed nor unemployed. The unemployment rate is defined on the labor force; changing the denominator changes the question, not the answer.*

**2. How much should you trust the second decimal?**

- c. The 95% confidence interval is roughly [3.98%, 4.24%]. Write the one-sentence version for the CEO of a staffing firm, with a minimum of statistical lingo.

*Solution: Something like: “Based on this survey, the unemployment rate in March 2019 was about 4.1%, and we can be quite confident it was between about 4.0% and 4.2%.”*

- d. Now ask your AI: “The BLS published an official unemployment rate for March 2019. Does your number match? Why or why not?” What does the gap teach you about that  $\pm 0.13$ -point interval?

*Solution: The published (seasonally adjusted) rate for March 2019 was 3.8%. The gap comes from things the naive calculation ignores: CPS survey weights (this extract is unweighted), the official 16-and-over universe, and seasonal adjustment. Lesson: the interval calculates sampling error only; definitional and design choices can move the number by more than the entire width of the CI.*

### **3. Decide whether to launch premium service**

Your staffing firm will launch a premium service this year only if average personal income among U.S. adults is at least \$41,500 — below that, the model says the price point fails. Ask your AI: “Compute average Total personal income and a 95% confidence interval.”

- e. Does the point estimate clear the bar? Does the confidence interval? What do you tell the CEO — go, no-go, or something else?

*Solution: Mean = \$41,754, 95% CI = [\$41,418, \$42,090] — the interval CONTAINS the \$41,500 threshold. The data cannot distinguish, and with  $n = 139,931$  already, “collect more data” probably isn’t going to help. The honest memo: the average is within noise of the threshold, so the launch decision must rest on the other economics, not on this estimate.*

- f. Before you sign that memo, ask the AI what is actually inside that average. Is the mean the right summary for this decision? One sentence.

*Solution: The “average adult income” quietly averages in 17,505 zeros, 156 negative incomes (mostly self-employed), and 147 incomes above \$1 million — a heavily right-skewed mix, so the mean sits well above the typical adult and may be the wrong summary entirely for a price-point decision aimed at typical customers (the median, or the share above a cutoff, may serve better).*

**Before you hand this in: add your team’s AI-use statement (a few sentences — which tools you used and what you used them for).**